

# Chirag Joshi

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## SUMMARY

AI / ML Engineer with over years of combined academic and applied experience in Large Language Models (LLMs), deep learning, and cloud-based AI systems. Experienced in designing, fine-tuning, and deploying production-ready AI applications using TensorFlow, PyTorch, and cloud AI services. Strong expertise in NLP, generative AI, multimodal learning, and scalable AI pipelines.

## EDUCATION

### University of Surrey

MSc, Artificial Intelligence

• **GPA:** Merit

- **Achievements:** Researched and build a Compatible, Low Parameters, High Performance Vision Language Model, Using a Novel Method Introduced by me, beating OpenAI Entry level Models with less Training.
- **Coursework:** Build and Fine Tuned Knife Classification System , Computer vision project, used to detect and classify knives for UK Police Department., Researched, Experimented and Documented Prompt Tuning Methodology for CLIP Models.

Sep 2024 - Sep 2025

Guildford, UK

### Shivaji University

BTech, Computer Science (Data Science)

• **GPA:** 8.3

- **Coursework:** Build a Library Management System using File Handling in C language., Build and deployed a Recommendation Engine, Song Recommendation System in Python., Build a Full Stack Web Application for Car Rental Services with AI Integrations.

Mar 2020 - Mar 2024

Maharashtra, India

## SKILLS

- **Languages:** Python, SQL
- **AI / ML :** PyTorch, Hugging Face, LLMs, LangGraph, RAG
- **Vector Databases:** FAISS
- **Backend / APIs:** FastAPI, Flask, Restful API, Redis, PostgreSQL
- **Cloud / Storage:** AWS EC2, AWS S3, AWS
- **DevOps / Observability:** Docker, GitHub, CI/CD
- **Frontend / Tools:** Streamlit, Git, HTML/CSS

## WORK EXPERIENCE

### Softtron Pvt. Ltd

AI /ML Developer Intern

Maharashtra, India · Jul 2023 - Aug 2023

- Built a prototype interview-screening assistant using Hugging Face models to score candidate responses, trigger follow-up logic, and summarise results into recruiter-facing feedback across 20 simulated interviews.
- Developed a document chunking and indexing pipeline for 300+ resumes, using AWS S3 for document storage and PostgreSQL for metadata; reduced lookup time to under 2 seconds per query in internal prototype testing.
- Deployed and tested application services on AWS EC2, and maintained a Postman collection covering edge cases, reducing manual regression time per release by 40%.
- Debugged data quality and model performance issues, demonstrating strong operational and troubleshooting skills.

## PROJECTS

### High Performance VLM Under 50M Parameters

- Designed a compact dual-encoder VLM using TinyViT + MiniLM with a Novel custom ConvFFN projection head.
- Implemented TinyCLIP-style distillation, multi-caption contrastive learning, and efficient batching strategies.
- Built distributed data loading and evaluation pipelines for MS-COCO and Flickr30k, computing Recall@K at scale.
- Achieved 42.3% Image to Text R@1 on COCO-5k and 56.1% on Flickr30k, outperforming OpenAI CLIP ViT-B/32 zero-shot baselines.
- Emphasized performance, scalability, and deployment feasibility, aligning with real-world system constraints.

### AI News Aggregator

- Architected a 5-stage AI pipeline: Data Ingestion → Content Processing → LLM Summarisation → Personalised Ranking → Email Delivery
- Integrated RSS feeds and YouTube API to collect structured and unstructured AI content from OpenAI, Anthropic, and technical channels.
- Converted URLs and video transcripts into structured knowledge representations using automated document processing workflows.
- Implemented LLM-based summarisation using GPT-4o-mini to generate concise, high-quality digests.
- Ensured production-ready quality with modular architecture, clean code practices, and deployment-ready configuration.

### AI Home Renovation Planner Agent

- Developed an AI Home Renovation Planner using Google ADK and Gemini 2.5 Flash with multimodal image understanding.
- Implemented a coordinator/dispatcher multi-agent architecture with 3 specialist agents for visual assessment, design planning, and project coordination.
- Built 5 tool-driven capabilities: renovation cost estimation, timeline calculation, photorealistic rendering generation, rendering editing, and reference image management.
- Supported 4 workflow types: current-room analysis, inspiration-driven redesign, text-only planning, and iterative feedback refinement.
- Delivered end-to-end renovation outputs including budget-aware cost ranges, project timelines, contractor/action checklists, and photorealistic renderings.

## CERTIFICATION

- Microsoft Azure AI Fundamentals
- Accenture: Data Analytics & Visualization
- BCG GenAI
- SQL

## EXTRACURRICULAR

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### **Boston Consulting Group (BCG )**

#### *BCG GenAI Job Simulation on Forage*

- Completed a job simulation involving AI-powered financial chatbot development for BCG's GenAI Consulting team.
- Gained experience in Python programming, including the use of libraries such as pandas for data manipulation.
- Integrated and interpreted complex financial data from 10-K and 10-Q reports, employing rule-based logic to create a chatbot that provides user-friendly financial insights and analysis.

### **Accenture**

#### *Accenture North America Data Analytics and Visualization Job Simulation on Forage*

- Completed a simulation focused on advising a hypothetical social media client as a Data Analyst at Accenture
- Cleaned, modelled and analyzed 7 datasets to uncover insights into content trends to inform strategic decisions
- Prepared a PowerPoint deck and video presentation to communicate key insights for the client and internal stakeholders